

EXTRACTION CODEBOOK

Systematic Review: Multimodal Cyber Threat Intelligence Fusion for Healthcare Cybersecurity Risk Assessment

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Corresponding Author: Oluseun Akeredolu, University of Warwick

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1. OVERVIEW AND PURPOSE

This codebook documents the systematic data extraction process for the review "Multimodal Cyber Threat Intelligence Fusion for Healthcare Cybersecurity Risk Assessment." It was developed to ensure consistent, transparent, and reproducible extraction across all 43 included studies.

Extraction Team: Two independent reviewers (R1, R2)

Inter-rater Reliability: Cohen's κ = 0.72-0.79 across coding stages

Conflict Resolution: Structured discussion + Delphi validation with NHS expert panel

2. EXTRACTION FRAMEWORK

The extraction schema is organized around six core taxonomy dimensions that capture the technical, regulatory, and operational aspects of CTI-CSRA integration:

Dimension	Focus Area	Key Questions
1. Methodological Paradigm	Technical approach	What primary methods are used?
2. Fusion Architecture	Integration strategy	How are multiple data types combined?
3. Healthcare Specificity	Clinical relevance	How applicable is this to healthcare?
4. Regulatory Embedding	Compliance focus	Are regulations explicitly addressed?
5. Validation Rigour	Evidence quality	How robust is the evaluation?
6. Operational Readiness	Deployment feasibility	Can this run in clinical settings?

3. DETAILED EXTRACTION FIELDS

3.1 Bibliographic Information

Field	Description	Format/Options	Example
study_id	Unique identifier for each study	S001, S002, ... S043	S001
authors	Complete author list	Full names as published	"Guo, Y., Liu, Z., Chen, X., et al."
year	Publication year	YYYY	2023
title	Full article title	Exact as published	"A framework for threat intelligence extraction and fusion"
source	Journal or conference	Full name	"Computers & Security"
doi	Digital Object Identifier	DOI string	"10.1016/j.cose.2023.103371"
url	URL if DOI not available	Full URL	" https://doi.org/10.1016/j.cose.2023.103371 "
study_type	Type of publication	Technical Framework, Empirical Evaluation, Systematic Review, Policy Analysis, Methodological Guide	Technical Framework

3.2 Methodological Paradigm

Definition: The primary technical approach or methodology employed for CTI processing and analysis.

Code	Paradigm	Description	Key Characteristics	Example Studies
GB	Graph-Based	Uses graph structures (knowledge graphs, graph neural networks, heterogeneous networks) to model relationships between threat entities	Relational reasoning, network analysis, entity linking, graph convolutional networks	Guo et al. (2023); Huang et al. (2024); Lin et al. (2023)
NLP	NLP-Centric	Focuses on natural language processing for	Text mining, entity recognition,	Silvestri et al. (2023); Xiao et al.

		extracting intelligence from unstructured text	sentiment analysis, semantic parsing, transformer models	(2024); Huang et al. (2024)
KB	Knowledge-Based	Relies on formal ontologies, rule-based systems, or explicit knowledge representations	Ontology-driven, semantic reasoning, rule engines, expert systems	Ampel et al. (2024); Samtani et al. (2022); Ren et al. (2023)
ES	Ensemble/Statistical	Combines multiple models or uses statistical methods for fusion and prediction	Multiple model integration, statistical fusion, feature combination, ensemble learning	Chakraborty et al. (2023); Powell & Drozdenko (2024); Negahdari Kia et al. (2023)

Coding Rules:

- If multiple paradigms are used, list primary first, then secondary
- For systematic reviews/surveys, code as "Not Applicable" with methodological note
- Base coding on explicit statements in methodology section

3.3 Fusion Architecture

Definition: The structural approach for integrating multiple data modalities or intelligence sources.

Code	Architecture	Description
DM	Deep Multimodal	End-to-end deep learning integration where multiple modalities are fused within the model architecture (e.g., attention-based fusion, cross-modal transformers)
MP	Modular Pipeline	Sequential or parallel processing pipelines where different modules handle different data types, with fusion at specific integration points
HY	Hybrid	Combines multiple fusion approaches or integrates different architectural patterns
NA	Not Applicable	Study does not implement multimodal fusion (e.g., single-modality analysis, purely theoretical)

Coding Rules:

- Document specific fusion mechanisms where described (e.g., "attention-based," "late fusion," "early fusion")
- Note if architecture is explicitly designed for clinical/healthcare data

3.4 Healthcare Specificity

Definition: The degree to which the study addresses or is applicable to healthcare contexts.

Code	Level	Description	Indicators
3	Very High	Direct healthcare application with clinical data/context	Uses IoMT/EHR data, clinical validation, hospital testbed, patient safety focus
2	High	Healthcare-focused but may use synthetic/simulated data	Healthcare use case, medical device focus, clinical scenarios
1	Generic	Mentions healthcare as potential application but not primary focus	Healthcare listed in discussion/future work, generic IT context
0	None	No healthcare relevance	Generic cybersecurity, critical infrastructure (non-healthcare)

Coding Rules:

- Look for explicit healthcare datasets (e.g., NHS, IoMT, clinical)
- Note specific clinical contexts (ICU, operating room, wearables, infusion pumps)
- Document if clinical experts were involved in validation

3.5 Regulatory Embedding

Definition: The extent to which healthcare regulations (GDPR, HIPAA, NHS DSPT) are explicitly addressed.

Code	Level	Description	Indicators
3	Clause-Level Enforcement	Explicit mapping to specific regulatory clauses with traceability	Cites specific articles/sections, demonstrates compliance mechanisms
2	Explicit Mapping	Regulations mentioned with clear relevance to the approach	Names specific regulations (GDPR, HIPAA, DSPT) with discussion
1	Nominal Mention	Regulations mentioned in passing without detailed integration	Generic compliance statements, regulations in introduction only
0	None	No regulatory discussion	Regulations not mentioned

Coding Rules:

- Record specific regulations mentioned (GDPR Art. 32, HIPAA §164.308, NHS DSPT)
- Note if compliance mechanisms are automated or manual
- Document any audit trail capabilities described

3.6 Validation Rigour

Definition: The robustness and realism of the evaluation approach.

Code	Level	Description	Indicators
4	NHS Operational Deployment	Validated in live clinical environment	NHS Trust pilot, real hospital deployment, ethics approval
3	Clinical Simulation	Healthcare-relevant simulation with realistic data	IoMT testbed, clinical scenario simulation, synthetic but realistic
2	Laboratory Testing	Controlled lab environment with benchmark datasets	Standard datasets (CIC-IDS, TON_IoT), academic testbed
1	Theoretical/Conceptual	No empirical validation	Framework proposal, conceptual design, literature review
0	None	No validation described	Purely theoretical

Coding Rules:

- Document specific datasets used (with DOIs/URLs where available)
- Note sample sizes, train/test splits, cross-validation
- Record performance metrics reported (accuracy, F1, precision, recall, latency, memory)

3.7 Operational Readiness

Definition: The feasibility of deploying the approach in resource-constrained clinical environments.

Code	Level	Description	Indicators
3	Edge-Deployable	Demonstrated on clinical-edge hardware with acceptable performance	Tested on Raspberry Pi, NVIDIA Jetson, ARM; latency <200ms; memory <100MB
2	Cloud/Server Ready	Runs on standard server/cloud infrastructure	GPU cluster, cloud server, high-performance computing
1	Conceptual Only	No deployment consideration	No discussion of hardware or performance constraints
0	Not Addressed	Deployment not mentioned	-

Coding Rules:

- Record reported latency (ms) and memory footprint (MB)
- Document hardware tested (specific models)
- Note any optimisation techniques (quantisation, pruning, distillation)
- Compare against clinical thresholds (from literature: <200ms, <100MB)

3.8 Thematic Coding (Grounded Theory)

Definition: Qualitative codes derived from open, axial, and selective coding of study findings and limitations.

Field	Description	Format
open_codes	Initial descriptive codes from first-pass reading	Comma-separated list
axial_codes	Grouped/categorized open codes	Comma-separated list
final_codes	Core themes emerging from selective coding	Comma-separated list
notes	Contextual observations, reviewer comments	Free text

Coding Rules:

- Two independent reviewers code each study
- Document disagreements and resolution
- Link codes to specific text/quotes from studies

Core Theme Definitions:

Final Code	Definition
Semantic Misalignment	Disconnect between technical CTI and clinical risk constructs
Regulatory Drift	Absence of clause-level regulatory traceability
Deployment Discontinuity	Inability to meet clinical edge-resource constraints
Methodological Opacity	Lack of transparency in methods, limiting reproducibility
Poor Generalizability	Models overfit to specific datasets, fail on novel data
Extraction of Implicit Threats	Difficulty inferring hidden threat semantics

3.9 Failure Mode Analysis

Definition: Identification and classification of documented limitations/failure modes.

Field	Description	Options
semantic_misalignment_present	Study exhibits semantic gap	Yes/No
regulatory_drift_present	Study lacks regulatory traceability	Yes/No
deployment_discontinuity_present	Study has deployment constraints	Yes/No
methodological_opacity_present	Study lacks transparency	Yes/No
severity_rating	Clinical impact severity	High/Medium/Low
patient_safety_impact	Documented safety implications	Free text

4. EXTRACTION PROCEDURE

4.1 Review Process

1. **Independent Extraction:** Both reviewers extract data independently for each study
2. **Comparison:** Extractions compared after each batch of 5-10 studies
3. **Disagreement Resolution:**
 - Initial discussion between reviewers
 - If unresolved, escalate to third reviewer
 - Delphi panel for clinical relevance disputes
4. **Reliability Monitoring:** Cohen's κ calculated at key stages

4.2 Quality Assurance

- **Calibration:** Initial training on 3 sample studies ($\kappa \geq 0.80$ required)
- **Audit Trail:** All extraction decisions documented
- **Version Control:** Extraction database versioned and archived
- **Transparency:** Complete extraction data available in supplementary materials

4.3 Data Recording

- **Primary Database:** studies.csv - Master list with all extracted fields
- **Thematic Database:** thematic_coding.csv - Qualitative coding data
- **Performance Database:** performance_metrics.csv - Quantitative metrics

- **Validation Database:** validation_rigour.csv - Validation characteristics
- **Regulatory Database:** regulatory_analysis.csv - Compliance analysis

5. CODING EXAMPLES

Example 1: Coding a Technical Framework Study

Study: Guo et al. (2023) - "A framework for threat intelligence extraction and fusion"

Field	Code	Justification
Methodological Paradigm	NLP-centric	Uses NLP for extraction from OSINT
Fusion Architecture	Modular pipeline	Sequential processing modules
Healthcare Specificity	Generic	No healthcare focus
Regulatory Embedding	None	No regulatory discussion
Validation Rigour	Benchmark	Tested on OSINT data, F1=81.37%
Operational Readiness	Cloud/Server Ready	Latency >300ms, high memory
Thematic Codes	Methodological Opacity, Deployment Discontinuity	High computational cost

Example 2: Coding a Healthcare-Focused Study

Study: Silvestri et al. (2023) - "A machine learning approach for the NLP-based analysis of cyber threats and vulnerabilities of the healthcare ecosystem"

Field	Code	Justification
Methodological Paradigm	NLP-centric	SecBERT for NER
Fusion Architecture	Modular pipeline	BERT-based with knowledge bases
Healthcare Specificity	High	EHR/IoMT focus
Regulatory Embedding	Nominal	Mentions GDPR only
Validation Rigour	Clinical Simulation	Healthcare dataset
Operational Readiness	Conceptual Only	No edge assessment
Thematic Codes	Regulatory Drift	Nominal regulatory embedding

6. REFERENCES TO MANUSCRIPT SECTIONS

Codebook Section	Corresponding Manuscript Section
Methodological Paradigm	Section 3.5, Table 4, Table 5
Fusion Architecture	Section 3.1, Table 4
Healthcare Specificity	Section 4, Table 4
Regulatory Embedding	Section 3.6, Section 4, Table 4
Validation Rigour	Section 3.7, Table 4
Operational Readiness	Section 3.3, Section 3.7, Table 4
Thematic Coding	Section 3.4, Section 3.8
Failure Mode Analysis	Section 3.8, Table 7

7. DATA AVAILABILITY

All extracted data is available at:

- **OSF Repository:** <https://doi.org/10.17605/OSF.IO/3CW8Y>
- **GitHub:** https://github.com/oluseun-akeredolu/HIMS-CDI/tree/review-v1.0/review_artefacts
- **Zenodo:** <https://doi.org/10.5281/zenodo.16897634>

Files Included:

- studies.csv - Complete extraction database
- thematic_coding.csv - Qualitative coding data
- performance_metrics.csv - Performance characteristics
- methodological_breakdown.csv - Paradigm analysis
- validation_rigour.csv - Validation assessment
- regulatory_analysis.csv - Regulatory compliance
- sensitivity_analysis.csv - Sensitivity analysis

8. VERSION HISTORY

Version	Date	Changes	Author
1.0	October 2024	Initial codebook for systematic review	OA